

Appendix

Figure 2.1A

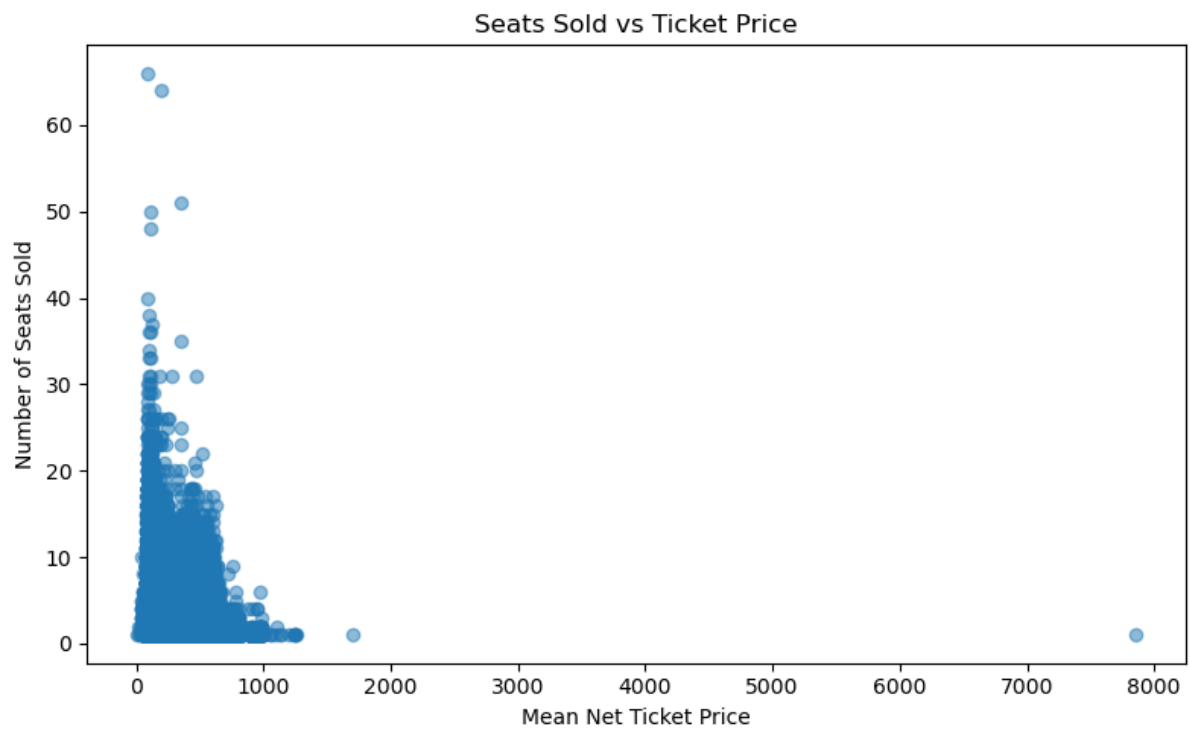


Figure 2.1B

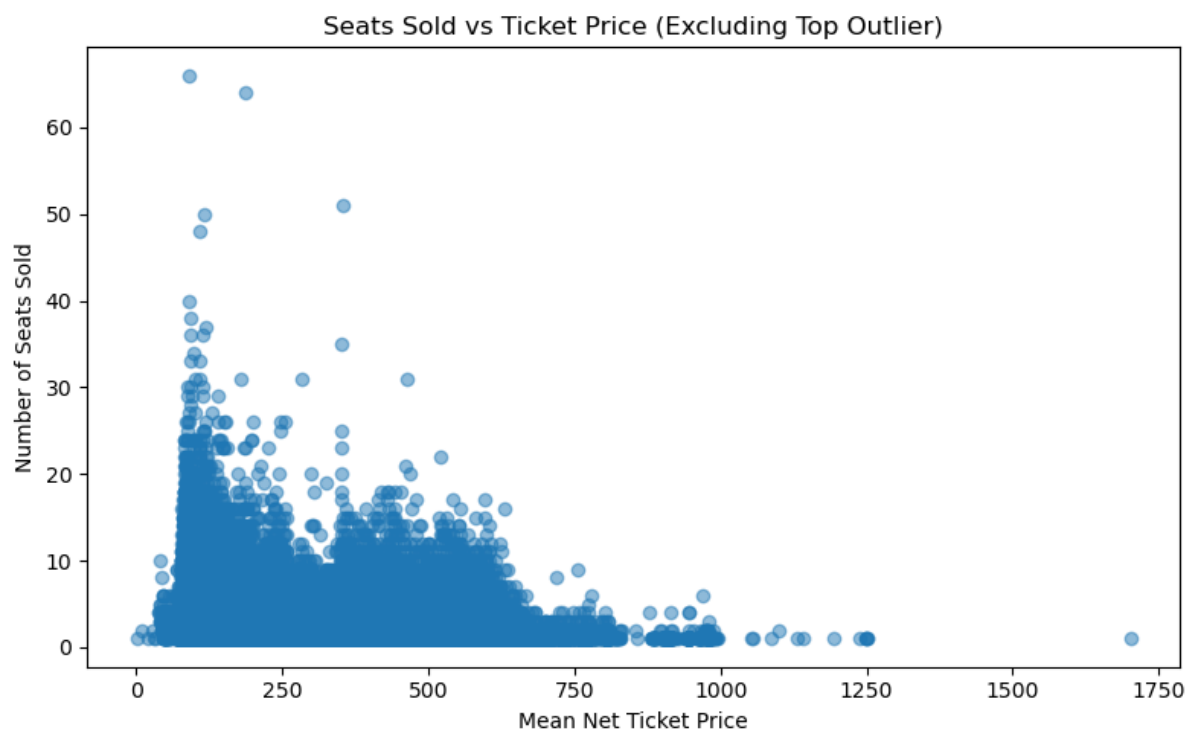


Figure 2.2

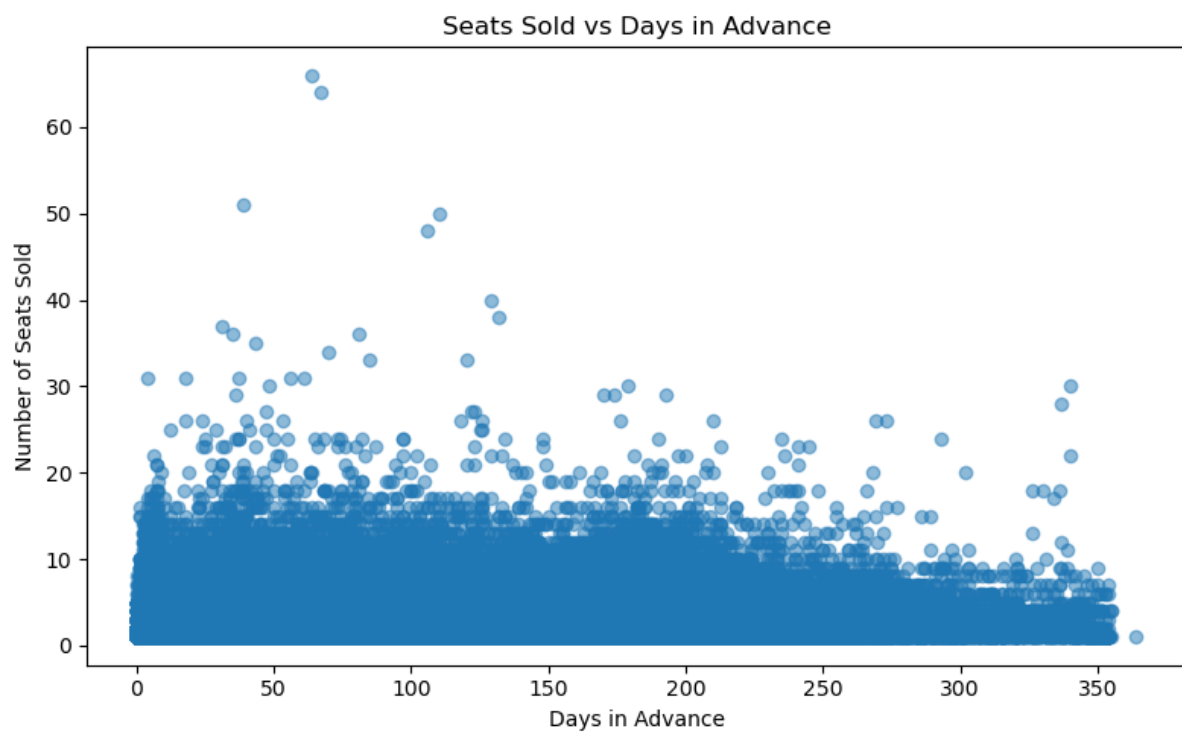


Figure 2.3A

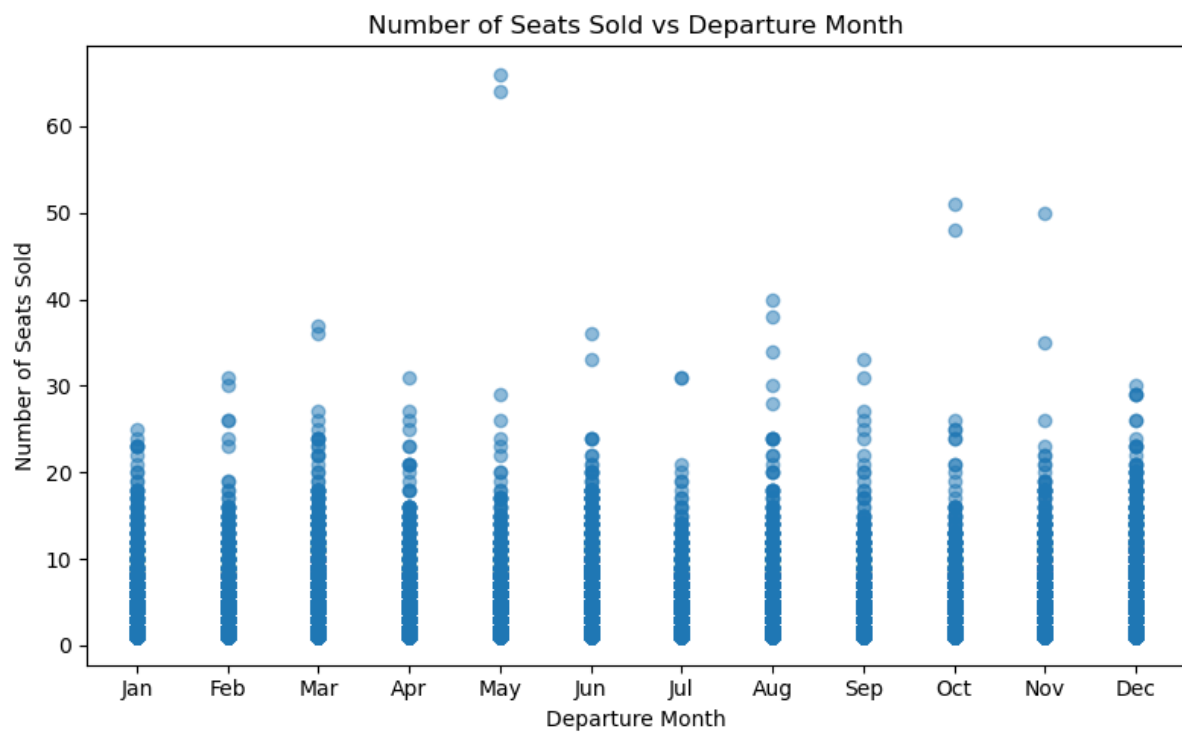


Figure 2.3B

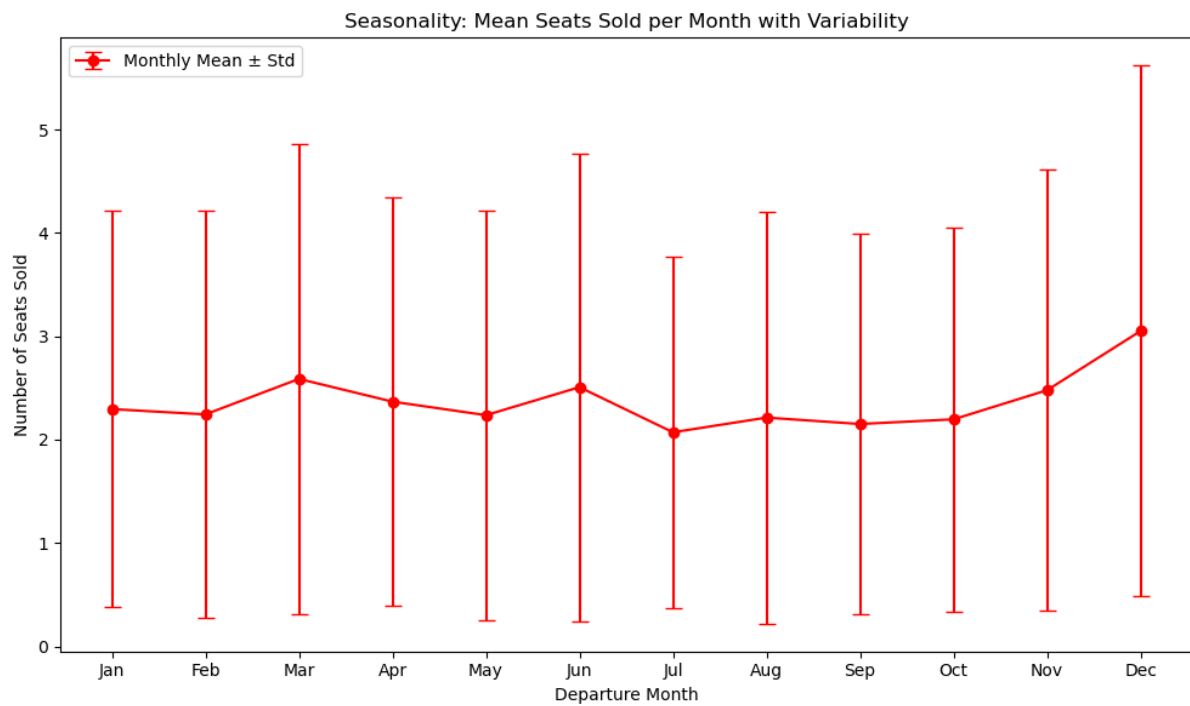


Figure 2.4

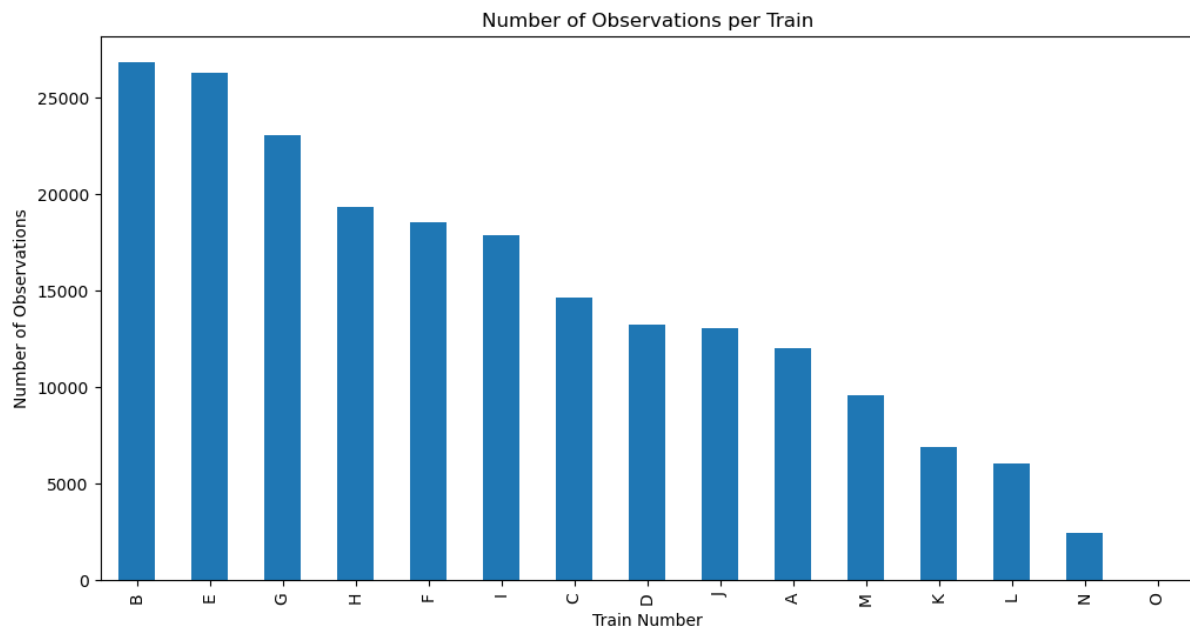


Figure 2.5

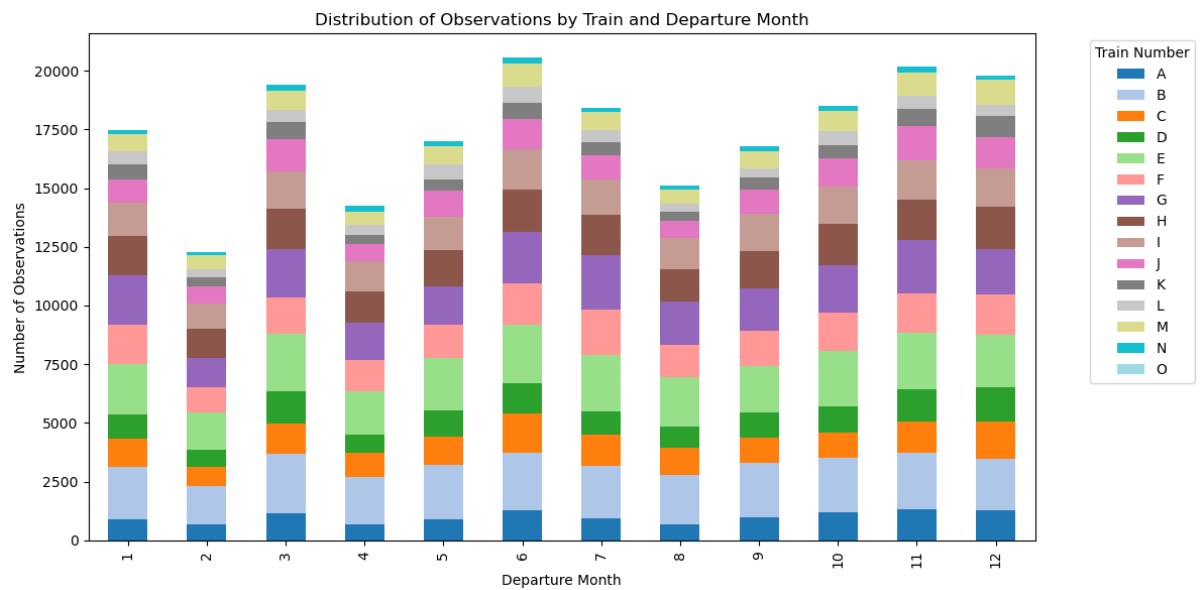


Table 3.1 - Identifying of outliers to be excluded using IQR

Q1 data

Q1: 4.690156280810966

Q3 data

Q3: 5.859102417486108

IQR: 1.1689461366751424

Lower bound

Lower bound: 2.936737075798252

Upper bound

Upper bound: 7.6125216224988215

Number of outliers: 3

Outlier records:

	mean_net_ticket_price	ln_P
45561	7855.766106	8.969003
160135	9.722006	2.274392
209366	1.278969	0.246054

Figure 4.1

```

=== Levels OLS (with Month FE, ticket_type baseline=Return_only) ===
                                OLS Regression Results
=====
Dep. Variable:          num_seats_total    R-squared:                0.097
Model:                  OLS               Adj. R-squared:         0.097
Method:                 Least Squares     F-statistic:             868.9
Date:                   Fri, 19 Sep 2025  Prob (F-statistic):       0.00
Time:                   22:27:07          Log-Likelihood:         -4.4075e+05
No. Observations:      209694            AIC:                   8.816e+05
Df Residuals:          209661            BIC:                   8.819e+05
Df Model:               32
Covariance Type:       HC3
=====
                                coef      std err          z      P>|z|      [0.025      0.975]
-----
Intercept                1.4392      0.030      48.132      0.000      1.381      1.498
C(ticket_type)[T.Oneway_only] -0.2668      0.017     -16.015      0.000     -0.299     -0.234
C(ticket_type)[T.Both]      -6.922e-05      0.025     -0.003      0.998     -0.049      0.048
C(ticket_type)[T.Neither]    0.0315      0.015      2.039      0.041      0.001      0.062
C(Dept_month)[T.2]          -0.0497      0.022     -2.246      0.025     -0.093     -0.006
C(Dept_month)[T.3]          0.3150      0.021     14.913      0.000      0.274      0.356
C(Dept_month)[T.4]          0.0890      0.021      4.157      0.000      0.047      0.131
C(Dept_month)[T.5]          0.0180      0.020      0.885      0.376     -0.022      0.058
C(Dept_month)[T.6]          0.3537      0.021     17.214      0.000      0.313      0.394
C(Dept_month)[T.7]          -0.1015      0.019     -5.434      0.000     -0.138     -0.065
C(Dept_month)[T.8]          -0.0353      0.021     -1.669      0.095     -0.077      0.006
C(Dept_month)[T.9]          -0.0776      0.020     -3.921      0.000     -0.116     -0.039
C(Dept_month)[T.10]         -0.0658      0.019     -3.409      0.001     -0.104     -0.028
C(Dept_month)[T.11]         0.1995      0.020     10.039      0.000      0.161      0.238
C(Dept_month)[T.12]         0.6663      0.022     29.929      0.000      0.623      0.710
C(Customer_Cat)[T.B]         0.7047      0.009     76.909      0.000      0.687      0.723
C(Train_Number_All)[T.B]     0.5763      0.024     24.226      0.000      0.530      0.623
C(Train_Number_All)[T.C]     0.2136      0.025      8.651      0.000      0.165      0.262
C(Train_Number_All)[T.D]     0.0534      0.024      2.212      0.027      0.006      0.101
C(Train_Number_All)[T.E]     0.6161      0.021     29.294      0.000      0.575      0.657
C(Train_Number_All)[T.F]     0.3680      0.021     17.370      0.000      0.327      0.410
C(Train_Number_All)[T.G]     0.4190      0.023     17.940      0.000      0.373      0.465
C(Train_Number_All)[T.H]     0.4142      0.022     19.097      0.000      0.372      0.457
C(Train_Number_All)[T.I]     0.1567      0.023      6.763      0.000      0.111      0.202
C(Train_Number_All)[T.J]     0.0454      0.021      2.176      0.030      0.005      0.086
C(Train_Number_All)[T.K]     -0.0988      0.026     -3.845      0.000     -0.149     -0.048
C(Train_Number_All)[T.L]     -0.0091      0.023     -0.394      0.693     -0.055      0.036
C(Train_Number_All)[T.M]     -0.2506      0.021    -11.832      0.000     -0.292     -0.209
C(Train_Number_All)[T.N]     -0.3586      0.025    -14.391      0.000     -0.407     -0.310
C(Train_Number_All)[T.O]     -1.2836      235.875     -0.005      0.996    -463.590     461.023
mean_net_ticket_price        -0.0008      4.61e-05    -18.181      0.000     -0.001     -0.001
days_in_advance              0.0025      9.14e-05     27.188      0.000      0.002      0.003
isNormCabin                  0.0539      0.012      4.436      0.000      0.030      0.078
=====
Omnibus:                 168526.069    Durbin-Watson:           1.821
Prob(Omnibus):           0.000    Jarque-Bera (JB):        9398972.015
Skew:                    3.465    Prob(JB):                0.00
Kurtosis:                35.058    Cond. No.                 1.26e+05
=====

```

Notes:

- [1] Standard Errors are heteroscedasticity robust (HC3)
- [2] The condition number is large, 1.26e+05. This might indicate that there are strong multicollinearity or other numerical problems.

Figure 4.2

=== Log-log OLS (with Month FE, ticket_type baseline=Return_only) ===

OLS Regression Results

```

Dep. Variable:          ln_Q      R-squared:          0.147
Model:                  OLS       Adj. R-squared:      0.147
Method:                  Least Squares      F-statistic:      1343.
Date:                    Fri, 19 Sep 2025    Prob (F-statistic): 0.00
Time:                    22:27:15           Log-Likelihood:   -1.9418e+05
No. Observations:        209694            AIC:             3.884e+05
Df Residuals:            209661            BIC:             3.888e+05
Df Model:                 32
Covariance Type:         HC3

```

	coef	std err	z	P> z	[0.025	0.975]
Intercept	1.2944	0.024	53.745	0.000	1.247	1.342
C(ticket_type)[T.Oneway_only]	-0.1045	0.005	-19.938	0.000	-0.115	-0.094
C(ticket_type)[T.Both]	-0.0028	0.007	-0.378	0.705	-0.017	0.012
C(ticket_type)[T.Neither]	0.0065	0.005	1.426	0.154	-0.002	0.015
C(Dept_month)[T.2]	-0.0317	0.007	-4.510	0.000	-0.045	-0.018
C(Dept_month)[T.3]	0.0833	0.006	12.908	0.000	0.071	0.096
C(Dept_month)[T.4]	0.0375	0.007	5.530	0.000	0.024	0.051
C(Dept_month)[T.5]	-0.0070	0.006	-1.091	0.275	-0.019	0.006
C(Dept_month)[T.6]	0.0976	0.006	15.472	0.000	0.085	0.110
C(Dept_month)[T.7]	-0.0546	0.006	-8.770	0.000	-0.067	-0.042
C(Dept_month)[T.8]	-0.0277	0.007	-4.162	0.000	-0.041	-0.015
C(Dept_month)[T.9]	-0.0533	0.006	-8.259	0.000	-0.066	-0.041
C(Dept_month)[T.10]	-0.0349	0.006	-5.580	0.000	-0.047	-0.023
C(Dept_month)[T.11]	0.0651	0.006	10.407	0.000	0.053	0.077
C(Dept_month)[T.12]	0.2347	0.007	35.401	0.000	0.222	0.248
C(Customer_Cat)[T.B]	0.2716	0.003	80.408	0.000	0.265	0.278
C(Train_Number_All)[T.B]	0.1858	0.007	24.862	0.000	0.171	0.200
C(Train_Number_All)[T.C]	0.0963	0.008	11.878	0.000	0.080	0.112
C(Train_Number_All)[T.D]	0.0204	0.008	2.543	0.011	0.005	0.036
C(Train_Number_All)[T.E]	0.2004	0.007	29.742	0.000	0.187	0.214
C(Train_Number_All)[T.F]	0.1408	0.007	19.968	0.000	0.127	0.155
C(Train_Number_All)[T.G]	0.1416	0.008	18.758	0.000	0.127	0.156
C(Train_Number_All)[T.H]	0.1463	0.007	20.866	0.000	0.133	0.160
C(Train_Number_All)[T.I]	0.0676	0.008	8.786	0.000	0.053	0.083
C(Train_Number_All)[T.J]	0.0185	0.007	2.512	0.012	0.004	0.033
C(Train_Number_All)[T.K]	-0.0444	0.009	-4.930	0.000	-0.062	-0.027
C(Train_Number_All)[T.L]	0.0068	0.009	0.764	0.445	-0.011	0.024
C(Train_Number_All)[T.M]	-0.1027	0.008	-13.234	0.000	-0.118	-0.087
C(Train_Number_All)[T.N]	-0.1668	0.010	-17.066	0.000	-0.186	-0.148
C(Train_Number_All)[T.O]	-0.5551	19.333	-0.029	0.977	-38.448	37.338
ln_P	-0.1929	0.004	-50.026	0.000	-0.200	-0.185
days_in_advance	0.0005	2.75e-05	19.088	0.000	0.000	0.001
isNormCabin	-0.0501	0.004	-12.659	0.000	-0.058	-0.042

```

Omnibus:                  12628.372      Durbin-Watson:          1.780
Prob(Omnibus):             0.000      Jarque-Bera (JB):       15063.115
Skew:                       0.656      Prob(JB):               0.00
Kurtosis:                   2.938      Cond. No.               4.16e+04

```

Notes:

[1] Standard Errors are heteroscedasticity robust (HC3)

[2] The condition number is large, 4.16e+04. This might indicate that there are strong multicollinearity or other numerical problems.

Rows before: 209694

Rows in wdf (levels): 209694

Rows in wdf2 (logs): 209694

Figure 4.3

Feature	VIF
const	19.089396
mean_net_ticket_price	2.3767
days_in_advance	1.366986
isNormCabin	2.063416
isReturn	1.017875
isOneway	1.036846

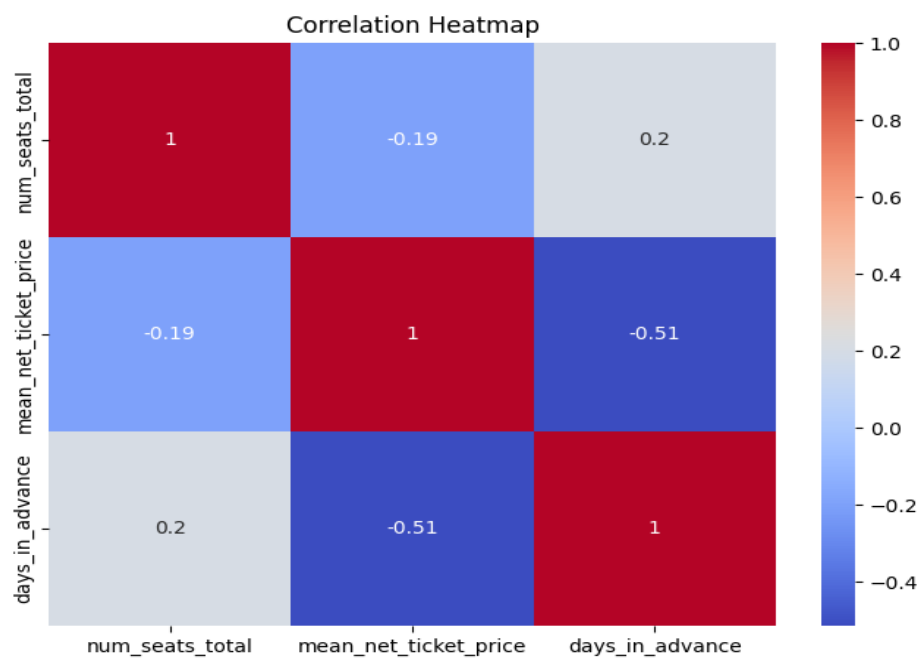


Figure 5.1A

First-stage (price on instrument):	
First Stage Estimation Results	
	mean_net_ticket_price
R-squared	0.6214
Partial R-squared	0.1155
Shea's R-squared	0.1155
Partial F-statistic	2.238e+04
P-value (Partial F-stat)	0.0000
Partial F-stat Distn	chi2(1)
=====	
Intercept	378.78
	(284.02)
isNormCabin	-156.29
	(-267.62)
C(ticket_type)[T.Oneway_only]	68.669
	(56.074)
C(ticket_type)[T.Both]	9.6064
	(8.7991)
C(ticket_type)[T.Neither]	10.282
	(13.358)
C(Dept_month)[T.2]	2.8984
	(2.6285)
C(Dept_month)[T.3]	-12.465
	(-13.530)
C(Dept_month)[T.4]	-0.9726
	(-0.9302)
C(Dept_month)[T.5]	-16.405
	(-17.124)
C(Dept_month)[T.6]	4.0344
	(4.2619)
C(Dept_month)[T.7]	-20.136
	(-21.682)
C(Dept_month)[T.8]	0.6936
	(0.6751)
C(Dept_month)[T.9]	-17.137
	(-17.959)
C(Dept_month)[T.10]	-11.710
	(-12.733)
C(Dept_month)[T.11]	10.279
	(11.354)
C(Dept_month)[T.12]	49.805
	(51.430)
C(Customer_Cat)[T.B]	-47.551
	(-76.799)
C(Train_Number_All)[T.B]	14.951
	(12.424)
C(Train_Number_All)[T.C]	46.031
	(33.654)
C(Train_Number_All)[T.D]	20.417
	(14.914)
C(Train_Number_All)[T.E]	-9.9394
	(-9.8563)
C(Train_Number_All)[T.F]	-0.3751
	(-0.3508)
C(Train_Number_All)[T.G]	11.305
	(9.0722)
C(Train_Number_All)[T.H]	-4.8621
	(-4.5690)
C(Train_Number_All)[T.I]	18.054
	(14.143)
C(Train_Number_All)[T.J]	-4.4156
	(-3.7629)
C(Train_Number_All)[T.K]	18.866
	(11.737)
C(Train_Number_All)[T.L]	29.887
	(18.904)
C(Train_Number_All)[T.M]	-8.6192
	(-6.7684)
C(Train_Number_All)[T.N]	26.836
	(11.887)
C(Train_Number_All)[T.O]	275.41
	(196.50)
days_in_advance	-0.5868
	(-149.61)

T-stats reported in parentheses

T-stats use same covariance type as original model

Figure 5.1B

```

Endogeneity tests (Wu-Hausman):
<bound method IVResults.wu_hausman of

=====
Dep. Variable:      num_seats_total    R-squared:      0.0636
Estimator:          IV-2SLS           Adj. R-squared:  0.0635
No. Observations:   209694           F-statistic:     1.003e+05
Date:               Fri, Sep 19 2025   P-value (F-stat) 0.0000
Time:               22:33:52           Distribution:     chi2(31)
Cov. Estimator:     robust

=====
                        Parameter Estimates
=====

```

	Parameter	Std. Err.	T-stat	P-value	Lower CI	Upper CI
Intercept	3.0440	0.0590	51.617	0.0000	2.9284	3.1596
isNormCabin	-0.6082	0.0275	-22.124	0.0000	-0.6621	-0.5544
C(ticket_type)[T.Oneway_only]	0.0241	0.0205	1.1746	0.2402	-0.0161	0.0643
C(ticket_type)[T.Both]	0.0406	0.0249	1.6293	0.1032	-0.0082	0.0895
C(ticket_type)[T.Neither]	0.0750	0.0158	4.7491	0.0000	0.0441	0.1060
C(Dept_month)[T.2]	-0.0374	0.0224	-1.6715	0.0946	-0.0813	0.0065
C(Dept_month)[T.3]	0.2622	0.0214	12.254	0.0000	0.2202	0.3041
C(Dept_month)[T.4]	0.0849	0.0218	3.8938	0.0001	0.0422	0.1276
C(Dept_month)[T.5]	-0.0515	0.0206	-2.5010	0.0124	-0.0919	-0.0111
C(Dept_month)[T.6]	0.3708	0.0209	17.735	0.0000	0.3298	0.4117
C(Dept_month)[T.7]	-0.1868	0.0189	-9.8667	0.0000	-0.2239	-0.1497
C(Dept_month)[T.8]	-0.0324	0.0217	-1.4898	0.1363	-0.0750	0.0102
C(Dept_month)[T.9]	-0.1502	0.0202	-7.4500	0.0000	-0.1897	-0.1107
C(Dept_month)[T.10]	-0.1154	0.0196	-5.8854	0.0000	-0.1538	-0.0770
C(Dept_month)[T.11]	0.2430	0.0203	11.982	0.0000	0.2033	0.2828
C(Dept_month)[T.12]	0.8773	0.0232	37.735	0.0000	0.8317	0.9229
C(Customer_Cat)[T.B]	0.5032	0.0133	37.903	0.0000	0.4772	0.5292
C(Train_Number_All)[T.B]	0.6397	0.0242	26.402	0.0000	0.5922	0.6871
C(Train_Number_All)[T.C]	0.4086	0.0265	15.431	0.0000	0.3567	0.4605
C(Train_Number_All)[T.D]	0.1399	0.0250	5.5949	0.0000	0.0909	0.1889
C(Train_Number_All)[T.E]	0.5740	0.0217	26.447	0.0000	0.5314	0.6165
C(Train_Number_All)[T.F]	0.3665	0.0218	16.771	0.0000	0.3236	0.4093
C(Train_Number_All)[T.G]	0.4668	0.0240	19.474	0.0000	0.4199	0.5138
C(Train_Number_All)[T.H]	0.3936	0.0223	17.680	0.0000	0.3500	0.4373
C(Train_Number_All)[T.I]	0.2332	0.0238	9.7812	0.0000	0.1864	0.2799
C(Train_Number_All)[T.J]	0.0267	0.0216	1.2346	0.2170	-0.0157	0.0691
C(Train_Number_All)[T.K]	-0.0189	0.0269	-0.7005	0.4836	-0.0716	0.0339
C(Train_Number_All)[T.L]	0.1175	0.0252	4.6544	0.0000	0.0680	0.1669
C(Train_Number_All)[T.M]	-0.2871	0.0220	-13.034	0.0000	-0.3303	-0.2439
C(Train_Number_All)[T.N]	-0.2449	0.0274	-8.9399	0.0000	-0.2986	-0.1912
C(Train_Number_All)[T.O]	-0.1168	0.0529	-2.2073	0.0273	-0.2205	-0.0131
mean_net_ticket_price	-0.0051	0.0001	-34.447	0.0000	-0.0054	-0.0048

```

=====
Endogenous: mean_net_ticket_price
Instruments: days_in_advance
Robust Covariance (Heteroskedastic)
Debiased: False
IVResults. id: 0x27a5181e300>

```

Figure 5.1C

```

=== 2SLS (LEVELS) ===
                                IV-2SLS Estimation Summary
=====
Dep. Variable:    num_seats_total    R-squared:    0.0636
Estimator:       IV-2SLS            Adj. R-squared: 0.0635
No. Observations: 209694            F-statistic:   1.003e+05
Date:            Fri, Sep 19 2025    P-value (F-stat) 0.0000
Time:            22:33:52            Distribution:   chi2(31)
Cov. Estimator:  robust

                                Parameter Estimates
=====

```

	Parameter	Std. Err.	T-stat	P-value	Lower CI	Upper CI
Intercept	3.0440	0.0590	51.617	0.0000	2.9284	3.1596
isNormCabin	-0.6082	0.0275	-22.124	0.0000	-0.6621	-0.5544
C(ticket_type)[T.Oneway_only]	0.0241	0.0205	1.1746	0.2402	-0.0161	0.0643
C(ticket_type)[T.Both]	0.0406	0.0249	1.6293	0.1032	-0.0082	0.0895
C(ticket_type)[T.Neither]	0.0750	0.0158	4.7491	0.0000	0.0441	0.1060
C(Dept_month)[T.2]	-0.0374	0.0224	-1.6715	0.0946	-0.0813	0.0065
C(Dept_month)[T.3]	0.2622	0.0214	12.254	0.0000	0.2202	0.3041
C(Dept_month)[T.4]	0.0849	0.0218	3.8938	0.0001	0.0422	0.1276
C(Dept_month)[T.5]	-0.0515	0.0206	-2.5010	0.0124	-0.0919	-0.0111
C(Dept_month)[T.6]	0.3708	0.0209	17.735	0.0000	0.3298	0.4117
C(Dept_month)[T.7]	-0.1868	0.0189	-9.8667	0.0000	-0.2239	-0.1497
C(Dept_month)[T.8]	-0.0324	0.0217	-1.4898	0.1363	-0.0750	0.0102
C(Dept_month)[T.9]	-0.1502	0.0202	-7.4500	0.0000	-0.1897	-0.1107
C(Dept_month)[T.10]	-0.1154	0.0196	-5.8854	0.0000	-0.1538	-0.0770
C(Dept_month)[T.11]	0.2430	0.0203	11.982	0.0000	0.2033	0.2828
C(Dept_month)[T.12]	0.8773	0.0232	37.735	0.0000	0.8317	0.9229
C(Customer_Cat)[T.B]	0.5032	0.0133	37.903	0.0000	0.4772	0.5292
C(Train_Number_All)[T.B]	0.6397	0.0242	26.402	0.0000	0.5922	0.6871
C(Train_Number_All)[T.C]	0.4086	0.0265	15.431	0.0000	0.3567	0.4605
C(Train_Number_All)[T.D]	0.1399	0.0250	5.5949	0.0000	0.0909	0.1889
C(Train_Number_All)[T.E]	0.5740	0.0217	26.447	0.0000	0.5314	0.6165
C(Train_Number_All)[T.F]	0.3665	0.0218	16.771	0.0000	0.3236	0.4093
C(Train_Number_All)[T.G]	0.4668	0.0240	19.474	0.0000	0.4199	0.5138
C(Train_Number_All)[T.H]	0.3936	0.0223	17.680	0.0000	0.3500	0.4373
C(Train_Number_All)[T.I]	0.2332	0.0238	9.7812	0.0000	0.1864	0.2799
C(Train_Number_All)[T.J]	0.0267	0.0216	1.2346	0.2170	-0.0157	0.0691
C(Train_Number_All)[T.K]	-0.0189	0.0269	-0.7005	0.4836	-0.0716	0.0339
C(Train_Number_All)[T.L]	0.1175	0.0252	4.6544	0.0000	0.0680	0.1669
C(Train_Number_All)[T.M]	-0.2871	0.0220	-13.034	0.0000	-0.3303	-0.2439
C(Train_Number_All)[T.N]	-0.2449	0.0274	-8.9399	0.0000	-0.2986	-0.1912
C(Train_Number_All)[T.O]	-0.1168	0.0529	-2.2073	0.0273	-0.2205	-0.0131
mean_net_ticket_price	-0.0051	0.0001	-34.447	0.0000	-0.0054	-0.0048

```

=====
Endogenous: mean_net_ticket_price
Instruments: days_in_advance
Robust Covariance (Heteroskedastic)
Debiased: False

```

Figure 5.2A

First-stage (ln_P on instrument):

First Stage Estimation Results

	ln_P
R-squared	0.6556
Partial R-squared	0.1669
Shea's R-squared	0.1669
Partial F-statistic	2.66e+04
P-value (Partial F-stat)	0.0000
Partial F-stat DISTR	chi2(1)
Intercept	5.9213 (1159.8)
isNormCabin	-0.6229 (-264.01)
C(ticket_type)[T.Oneway_only]	0.1856 (50.055)
C(ticket_type)[T.Both]	0.0304 (6.7357)
C(ticket_type)[T.Neither]	0.0193 (6.6839)
C(Dept_month)[T.2]	0.0040 (0.8808)
C(Dept_month)[T.3]	-0.0742 (-19.325)
C(Dept_month)[T.4]	-0.0123 (-2.9449)
C(Dept_month)[T.5]	-0.0729 (-18.786)
C(Dept_month)[T.6]	0.0140 (3.8117)
C(Dept_month)[T.7]	-0.0925 (-24.526)
C(Dept_month)[T.8]	-0.0132 (-3.1800)
C(Dept_month)[T.9]	-0.1035 (-26.559)
C(Dept_month)[T.10]	-0.0640 (-17.012)
C(Dept_month)[T.11]	0.0550 (15.120)
C(Dept_month)[T.12]	0.2600 (67.779)
C(Customer_Cat)[T.B]	-0.2168 (-104.13)
C(Train_Number_All)[T.B]	0.0411 (8.8167)
C(Train_Number_All)[T.C]	0.2318 (47.791)
C(Train_Number_All)[T.D]	0.0613 (11.816)
C(Train_Number_All)[T.E]	-0.0397 (-9.7722)
C(Train_Number_All)[T.F]	0.0094 (2.1949)
C(Train_Number_All)[T.G]	0.0187 (3.9446)
C(Train_Number_All)[T.H]	-0.0240 (-5.5687)
C(Train_Number_All)[T.I]	0.0593 (12.056)
C(Train_Number_All)[T.J]	-0.0190 (-4.0427)
C(Train_Number_All)[T.K]	0.0649 (10.893)
C(Train_Number_All)[T.L]	0.1692 (32.193)
C(Train_Number_All)[T.M]	-0.0283 (-5.5119)
C(Train_Number_All)[T.N]	0.1471 (20.173)
C(Train_Number_All)[T.O]	0.5383 (103.65)
days_in_advance	-0.0029 (-163.10)

T-stats reported in parentheses

T-stats use same covariance type as original model

Figure 5.2B

```

Endogeneity tests (Wu-Hausman):
<bound method IVResults.wu_hausman of
=====
Dep. Variable:          ln_Q      R-squared:          0.1376
Estimator:             IV-2SLS   Adj. R-squared:     0.1374
No. Observations:      209694    F-statistic:        2.248e+05
Date:                  Fri, Sep 19 2025    P-value (F-stat)    0.0000
Time:                  22:34:09    Distribution:        chi2(31)
Cov. Estimator:        robust

=====
                        Parameter Estimates
=====

```

	Parameter	Std. Err.	T-stat	P-value	Lower CI	Upper CI
Intercept	2.3710	0.0512	46.290	0.0000	2.2706	2.4714
isNormCabin	-0.1633	0.0070	-23.489	0.0000	-0.1770	-0.1497
C(ticket_type)[T.Oneway_only]	-0.0707	0.0056	-12.712	0.0000	-0.0816	-0.0598
C(ticket_type)[T.Both]	0.0027	0.0074	0.3685	0.7125	-0.0118	0.0172
C(ticket_type)[T.Neither]	0.0100	0.0046	2.1821	0.0291	0.0010	0.0190
C(Dept_month)[T.2]	-0.0309	0.0070	-4.3930	0.0000	-0.0447	-0.0171
C(Dept_month)[T.3]	0.0698	0.0065	10.789	0.0000	0.0572	0.0825
C(Dept_month)[T.4]	0.0353	0.0068	5.1823	0.0000	0.0219	0.0486
C(Dept_month)[T.5]	-0.0202	0.0064	-3.1719	0.0015	-0.0327	-0.0077
C(Dept_month)[T.6]	0.1002	0.0063	15.851	0.0000	0.0878	0.1126
C(Dept_month)[T.7]	-0.0714	0.0062	-11.494	0.0000	-0.0836	-0.0592
C(Dept_month)[T.8]	-0.0301	0.0067	-4.4829	0.0000	-0.0433	-0.0170
C(Dept_month)[T.9]	-0.0721	0.0065	-11.123	0.0000	-0.0848	-0.0594
C(Dept_month)[T.10]	-0.0465	0.0063	-7.4109	0.0000	-0.0589	-0.0342
C(Dept_month)[T.11]	0.0752	0.0063	11.954	0.0000	0.0628	0.0875
C(Dept_month)[T.12]	0.2819	0.0067	42.002	0.0000	0.2688	0.2951
C(Customer_Cat)[T.B]	0.2322	0.0042	55.431	0.0000	0.2240	0.2404
C(Train_Number_All)[T.B]	0.1933	0.0075	25.742	0.0000	0.1786	0.2080
C(Train_Number_All)[T.C]	0.1384	0.0084	16.444	0.0000	0.1219	0.1549
C(Train_Number_All)[T.D]	0.0316	0.0081	3.8921	0.0001	0.0157	0.0475
C(Train_Number_All)[T.E]	0.1932	0.0068	28.263	0.0000	0.1798	0.2066
C(Train_Number_All)[T.F]	0.1425	0.0071	19.987	0.0000	0.1285	0.1565
C(Train_Number_All)[T.G]	0.1450	0.0076	19.071	0.0000	0.1301	0.1599
C(Train_Number_All)[T.H]	0.1419	0.0071	20.013	0.0000	0.1280	0.1558
C(Train_Number_All)[T.I]	0.0784	0.0078	10.113	0.0000	0.0632	0.0936
C(Train_Number_All)[T.J]	0.0151	0.0075	2.0151	0.0439	0.0004	0.0297
C(Train_Number_All)[T.K]	-0.0326	0.0091	-3.5678	0.0004	-0.0505	-0.0147
C(Train_Number_All)[T.L]	0.0375	0.0091	4.1083	0.0000	0.0196	0.0554
C(Train_Number_All)[T.M]	-0.1078	0.0079	-13.725	0.0000	-0.1232	-0.0924
C(Train_Number_All)[T.N]	-0.1401	0.0100	-13.944	0.0000	-0.1598	-0.1204
C(Train_Number_All)[T.O]	-0.4573	0.0105	-43.456	0.0000	-0.4779	-0.4366
ln_P	-0.3747	0.0086	-43.389	0.0000	-0.3916	-0.3578

```

=====
Endogenous: ln_P
Instruments: days_in_advance
Robust Covariance (Heteroskedastic)
Debiased: False
IVResults, id: 0x27a3cc16000>

```


Figure 5.2C

```

=== 2SLS (LOG-LOG) ===
                                IV-2SLS Estimation Summary
=====
Dep. Variable:                   ln_Q    R-squared:                   0.1376
Estimator:                       IV-2SLS  Adj. R-squared:           0.1374
No. Observations:                 209694  F-statistic:                 2.248e+05
Date:                            Fri, Sep 19 2025  P-value (F-stat)       0.0000
Time:                            22:34:09  Distribution:                 chi2(31)
Cov. Estimator:                   robust

                                Parameter Estimates
=====
Parameter      Std. Err.      T-stat      P-value      Lower CI      Upper CI
-----
Intercept      2.3710      0.0512      46.290      0.0000      2.2706      2.4714
isNormCabin    -0.1633     0.0070     -23.489     0.0000     -0.1770     -0.1497
C(ticket_type)[T.Oneway_only] -0.0707     0.0056     -12.712     0.0000     -0.0816     -0.0598
C(ticket_type)[T.Both]         0.0027     0.0074      0.3685     0.7125     -0.0118     0.0172
C(ticket_type)[T.Neither]      0.0100     0.0046      2.1821     0.0291      0.0010     0.0190
C(Dept_month)[T.2]             -0.0309     0.0070     -4.3930     0.0000     -0.0447     -0.0171
C(Dept_month)[T.3]             0.0698     0.0065     10.789     0.0000      0.0572     0.0825
C(Dept_month)[T.4]             0.0353     0.0068      5.1823     0.0000      0.0219     0.0486
C(Dept_month)[T.5]            -0.0202     0.0064     -3.1719     0.0015     -0.0327     -0.0077
C(Dept_month)[T.6]             0.1002     0.0063     15.851     0.0000      0.0878     0.1126
C(Dept_month)[T.7]            -0.0714     0.0062     -11.494     0.0000     -0.0836     -0.0592
C(Dept_month)[T.8]            -0.0301     0.0067     -4.4829     0.0000     -0.0433     -0.0170
C(Dept_month)[T.9]            -0.0721     0.0065     -11.123     0.0000     -0.0848     -0.0594
C(Dept_month)[T.10]           -0.0465     0.0063     -7.4109     0.0000     -0.0589     -0.0342
C(Dept_month)[T.11]           0.0752     0.0063     11.954     0.0000      0.0628     0.0875
C(Dept_month)[T.12]           0.2819     0.0067     42.002     0.0000      0.2688     0.2951
C(Customer_Cat)[T.B]          0.2322     0.0042     55.431     0.0000      0.2240     0.2404
C(Train_Number_All)[T.B]      0.1933     0.0075     25.742     0.0000      0.1786     0.2080
C(Train_Number_All)[T.C]      0.1384     0.0084     16.444     0.0000      0.1219     0.1549
C(Train_Number_All)[T.D]      0.0316     0.0081      3.8921     0.0001      0.0157     0.0475
C(Train_Number_All)[T.E]      0.1932     0.0068     28.263     0.0000      0.1798     0.2066
C(Train_Number_All)[T.F]      0.1425     0.0071     19.987     0.0000      0.1285     0.1565
C(Train_Number_All)[T.G]      0.1450     0.0076     19.071     0.0000      0.1301     0.1599
C(Train_Number_All)[T.H]      0.1419     0.0071     20.013     0.0000      0.1280     0.1558
C(Train_Number_All)[T.I]      0.0784     0.0078     10.113     0.0000      0.0632     0.0936
C(Train_Number_All)[T.J]      0.0151     0.0075      2.0151     0.0439      0.0004     0.0297
C(Train_Number_All)[T.K]     -0.0326     0.0091     -3.5678     0.0004     -0.0505     -0.0147
C(Train_Number_All)[T.L]      0.0375     0.0091      4.1083     0.0000      0.0196     0.0554
C(Train_Number_All)[T.M]     -0.1078     0.0079     -13.725     0.0000     -0.1232     -0.0924
C(Train_Number_All)[T.N]     -0.1401     0.0100     -13.944     0.0000     -0.1598     -0.1204
C(Train_Number_All)[T.O]     -0.4573     0.0105     -43.456     0.0000     -0.4779     -0.4366
ln_P             -0.3747     0.0086     -43.389     0.0000     -0.3916     -0.3578
=====

Endogenous: ln_P
Instruments: days_in_advance
Robust Covariance (Heteroskedastic)
Debiased: False

Price elasticity (2SLS): -0.3746807927525424

```